

Topic Module Computer Laboratory	Hydraulic Modelling	
	1.	Surface Water Modelling
	3	Calculation of the water surface profile upstream of a dam

One purpose for which dams are designed and built is flood protection. Among the range of criteria that need to be considered, the design engineer must ascertain that the dam height is sufficient for the annual exceedance probability of the design flood, and the design engineer must also ascertain that the change in water level upstream of the dam does not result in negative consequences. You will investigate these two aspects in detail. For the channel under consideration with trapezoidal geometry (base width $b = 10$ m, side slope $1:m = 1:2$ and channel bed slope $S_0 = 0.25\%$), compute the water surface profile for a design discharge $Q = 25$ m³/s and a Manning's roughness coefficient $n = 0.02857$ s/m^{1/3}. The flow is subcritical with a uniform flow depth $y_0 = 2.25$ m. The proposed dam blocks the water at $x = 0$ m up to an elevation $y_p = 382$ m above mean sea level with a water depth at the dam of $y = 3.10$ m.

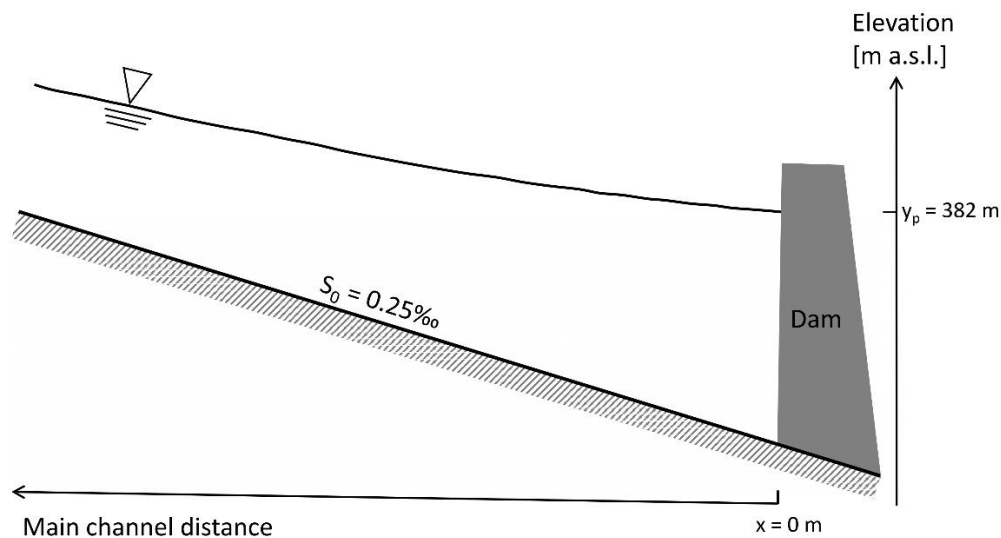


Figure 1a: Longitudinal-section view of the trapezoidal channel

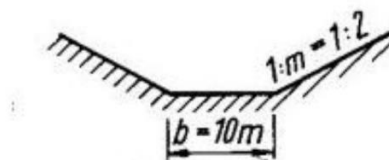


Figure 1b: Cross-section view of the trapezoidal channel

This activity includes

- Water surface profile calculation using the standard step method using HEC-RAS.
- Plotting of the resulting water surface profiles (i) as longitudinal and (ii) cross section (choose various cross-sections) plots.
- Exploring tabulated output and summary statistics.

Resources

- Computer software HEC-RAS. HEC-RAS can be obtained free of charge at <https://www.hec.usace.army.mil/software/hec-ras/>.

Instructions

Start the software

Starting HEC-RAS

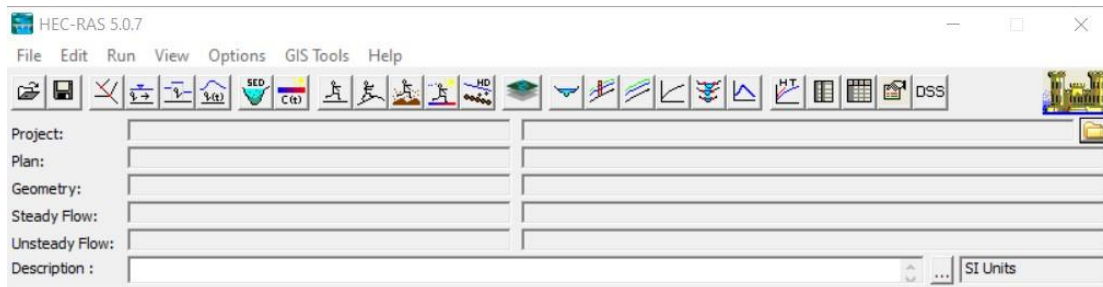


Figure 2: HEC-RAS user interface

Setting up projects

New project: **File – New Project ...**

In the working directory **Create Folder: WaterLevel**

Title: Water Level

File Name: WaterLevel.prj

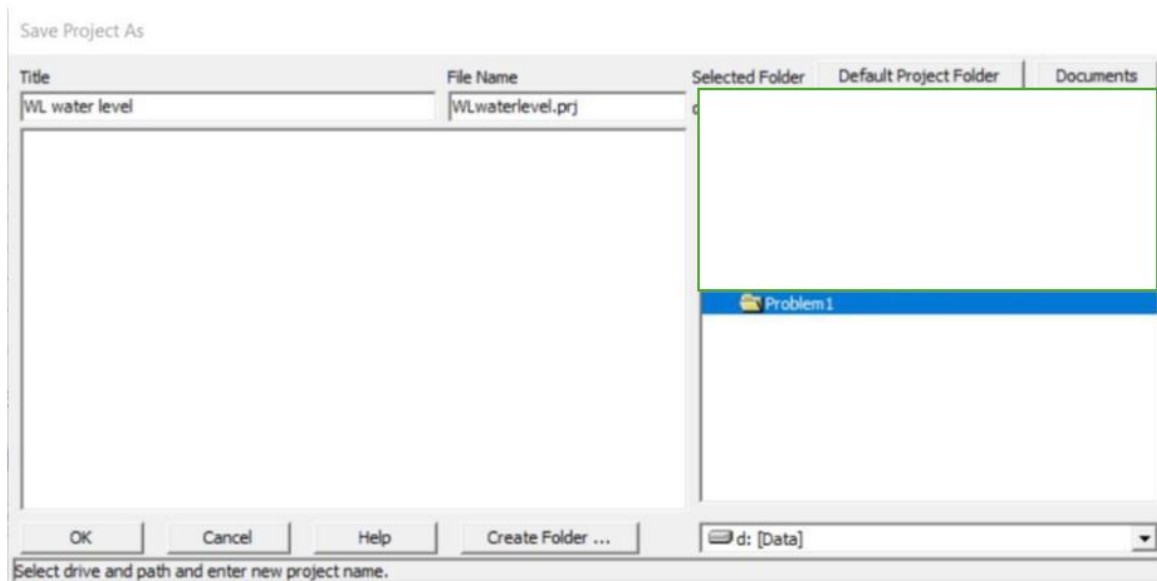


Figure 3: Setting up a new HEC-RAS project

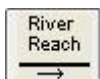
System of units

Select metric SI units system under: **Options – Unit system**
and activate **Set as default for new projects**

Geometry

Edit – Geometric Data

Click on the **River Reach** button to enter the river system schematic.



Shown is the top view. Always enter the river system in the direction of flow, upstream to downstream. Left mouse click begins the entry, a right mouse click deletes the last entry, left mouse double-clicking finalises the entry of the river system. Here a straight channel is considered, i.e. the beginning and end point, and possibly sections in between are required.

Next enter:

River: Channel

Reach: lower

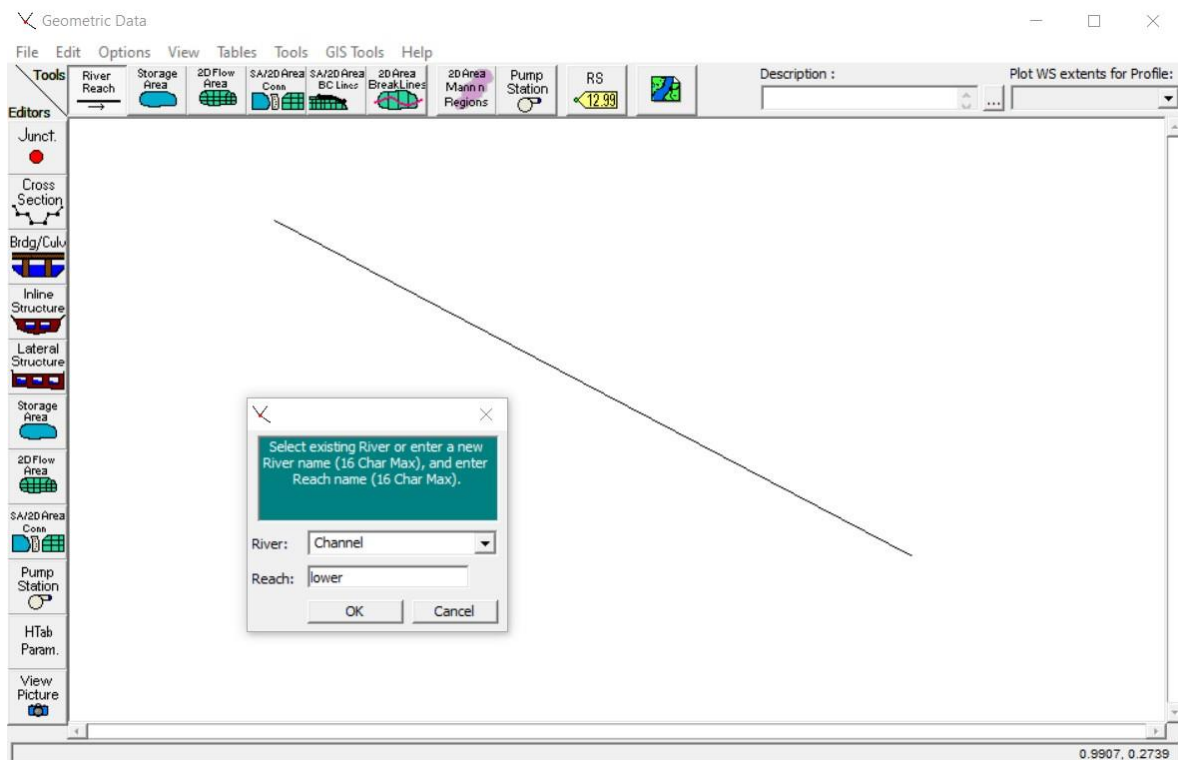
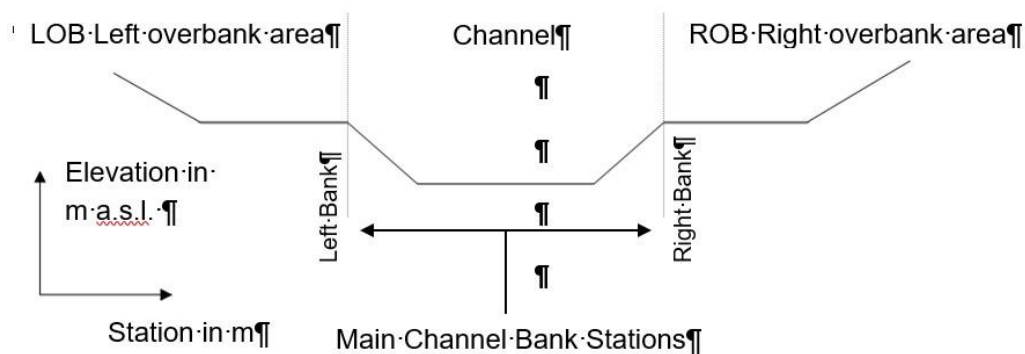


Figure 4: Entering the river system schematic

Cross sections:

Entering the cross section:



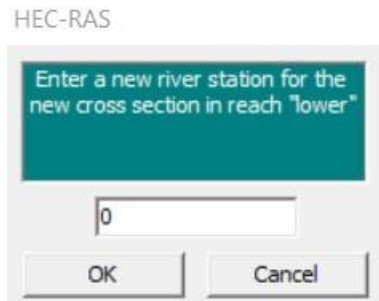
Button **Cross Section**



Options – Add a new Cross Section

Enter River Station: numerical value which decreases in the direction of flow, e.g. kilometre values (value is used to number/sort the River Stations)

Here: downstream cross section is **River station: 0**



Description: downstream boundary

Downstream Reach Length: 0

(since this is already the lower cross section, the length to the next section downstream is 0 m; the information for LOB and ROB are to be considered for any increased / reduced flow lengths in the area of the foreland (overbanks); here: no forelands (overbanks))

Cross Section X-Y Coordinates: Entry of the cross section from left (Station = 0) to right in the direction of flow

Use: Height of the cross-section profile = 4 m, elevation of the channel bottom = 378.9 m a.m.s.l

Mannings n Values: Separate entries for the main channel and the overbanks (here: since there are no overbanks in this case the value for LOB and ROB is of no significance; use: $n_{LOB} = n_{ROB} = n_{Channel} = 0.02857 \text{ s/m}^{1/3}$)

Main Channel Bank Stations: signifies the separation points between overbanks and main channel; here: enter the entire width (there are no overbanks in this case)

Apply Data: Saves the entries and displays the profile based on the data entered

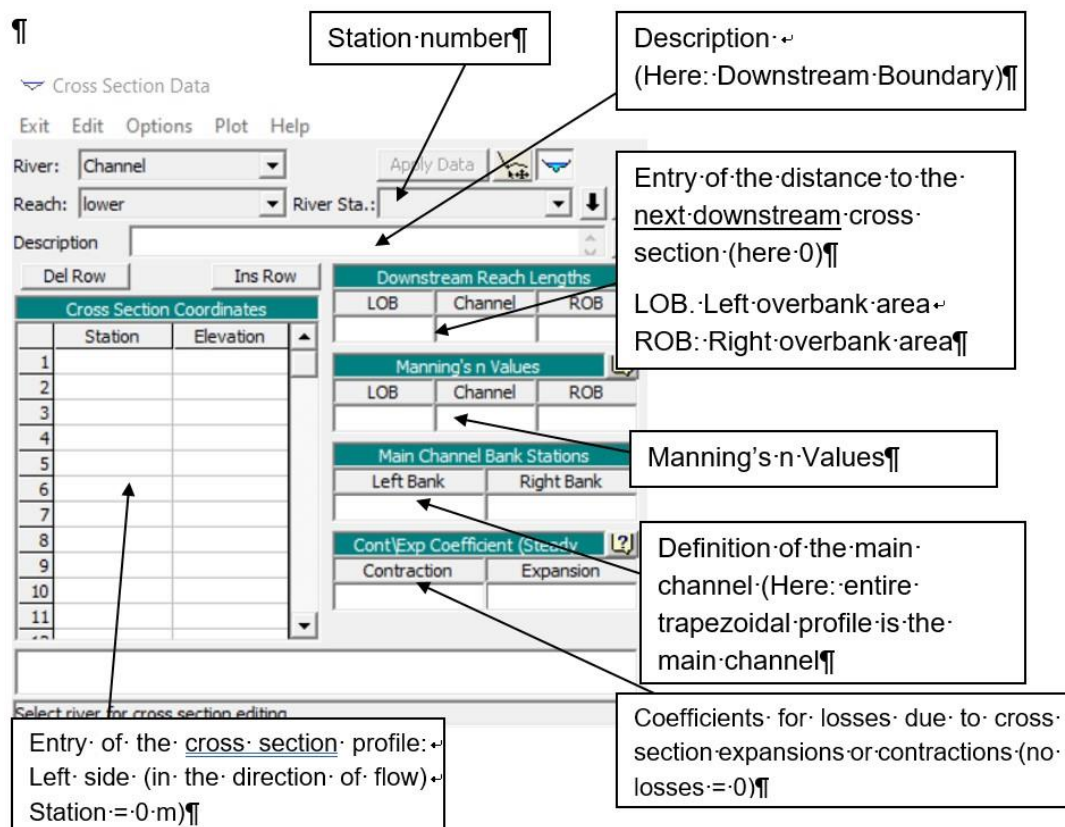


Figure 6: Cross section data entries

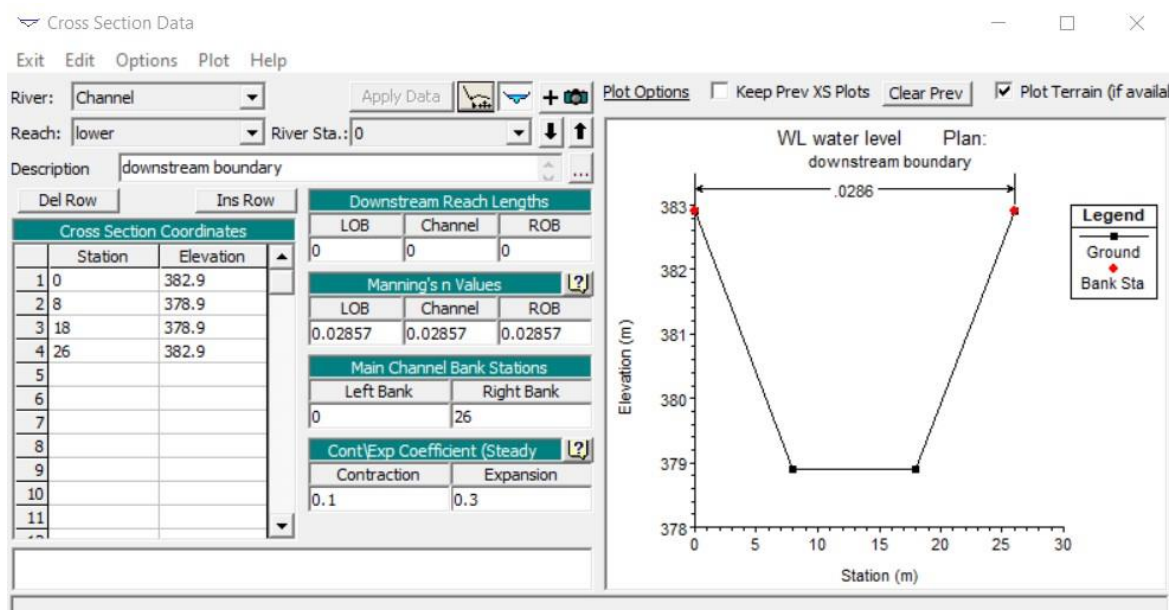


Figure 7: Cross section entries, River Station 0

Entries for the cross section located upstream:

Same geometry as the downstream cross section: Copy the cross-section data via **Options – Copy Current Cross Section**

Copy Cross Section

Select a River and Reach and then enter a new river station.

River: Channel

Reach: lower River Sta: 1

OK Cancel

River station: 1

Description: upstream boundary

Downstream Reach Length: 11,025 m

Adjustment of the elevation via: **Options – Adjust Elevations** (determine value using the slope: $(0.25/1,000) \times 11,025 \text{ m} = 2.75625 \text{ m}$; the change in elevation is entered as positive number, since River Station 1 is located upstream:

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Enter amount to add to Elevations (+/-)

2.75625

OK Cancel

Figure 8: Adjustment of the elevation for River Station 1

Apply Data: Saves the entries and displays the profile based on the data entered

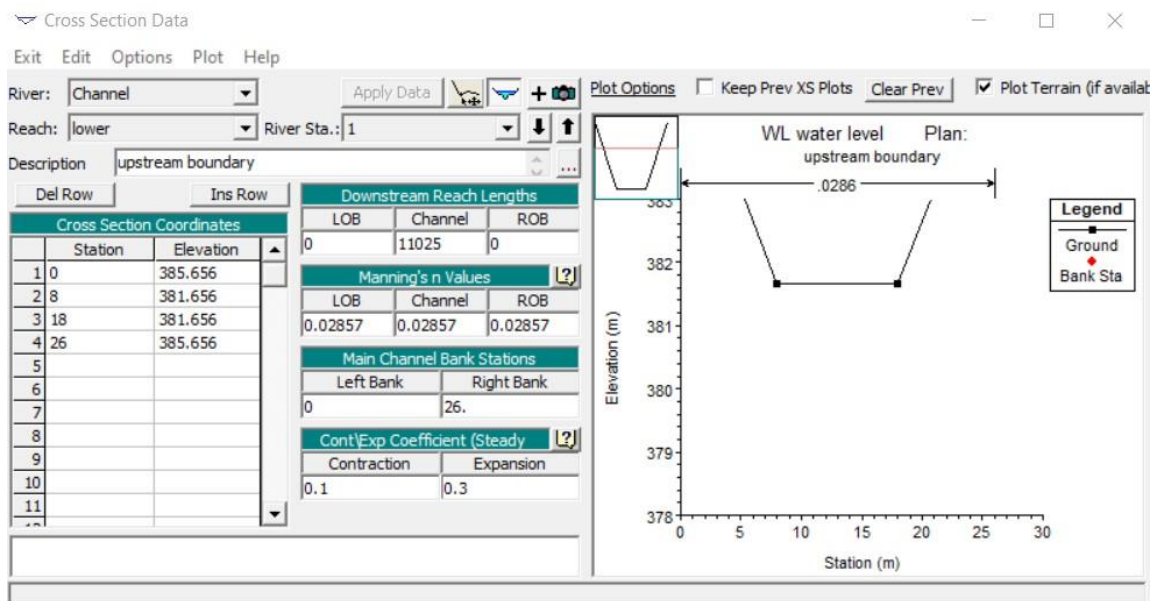


Figure 9: Cross-section entries, River Station 1

Exit – Exit Cross Section Editor

Save the geometry data: **File – Save Geometry Data As ...**
(Title: WL channel geometry)

File – Exit Geometry Data Editor

Inputs for the computations: Flow Data

Steady flow: **Edit – Steady Flow Data**

Entry of the discharge (Flow Change Locations)

Here: **Q = 25 m³/s** (Entry for PF 1)

Steady Flow Data

File Options Help

Description : Q

Enter/Edit Number of Profiles (32000 max): 1 Reach Boundary Conditions ...

Locations of Flow Data Changes

River: Channel Add Multiple...

Reach: lower River Sta.: 1 Add A Flow Change Location

Flow Change Location				Profile Names and Flow Rates	
	River	Reach	RS	PF 1	
1	Channel	lower	1	25	

Edit Steady flow data for the profiles (m3/s)

Figure 10: Input of discharge

Boundary conditions:

Subcritical flow è downstream boundary condition required

Button **Reach Boundary Conditions...**

Steady Flow Boundary Conditions

☒ Set boundary for all profiles ☐ Set boundary for one profile at a time

Available External Boundary Condition Types

Known W.S. Critical Depth Normal Depth Rating Curve Delete

Selected Boundary Condition Locations and Types

River	Reach	Profile	Upstream	Downstream
Channel	lower	all		

Steady Flow Reach-Storage Area Optimization ...

OK Cancel Help

Enter to make the boundary for selected location a known water surface.

Figure 11: Input of the boundary conditions

Boundary condition: Enter **Known W.S.** (see problem description, enter values in m a.s.l.)
 Known W.S. is a known water level, in this case at River Station 0 (Downstream). See Figure 1: 382 m a.s.l.

HEC-RAS

Set known water surfaces for flows.

	Flow (m ³ /s)	Known WS El (m)
1	25	382

OK Cancel

Figure 12: Input of known (downstream) water level

Steady Flow Boundary Conditions – OK

Button: **Apply Data**

File – Save Flow Data As ...

(Title: WL Flow Data)

File – Exit Flow Data Editor

Calculations

Run – Steady Flow Analysis

File - New Plan

(Title: WL; button **OK**; **Short Plan Identifier**: Calc 1)

Subcritical flow **Flow Regime**: subcritical

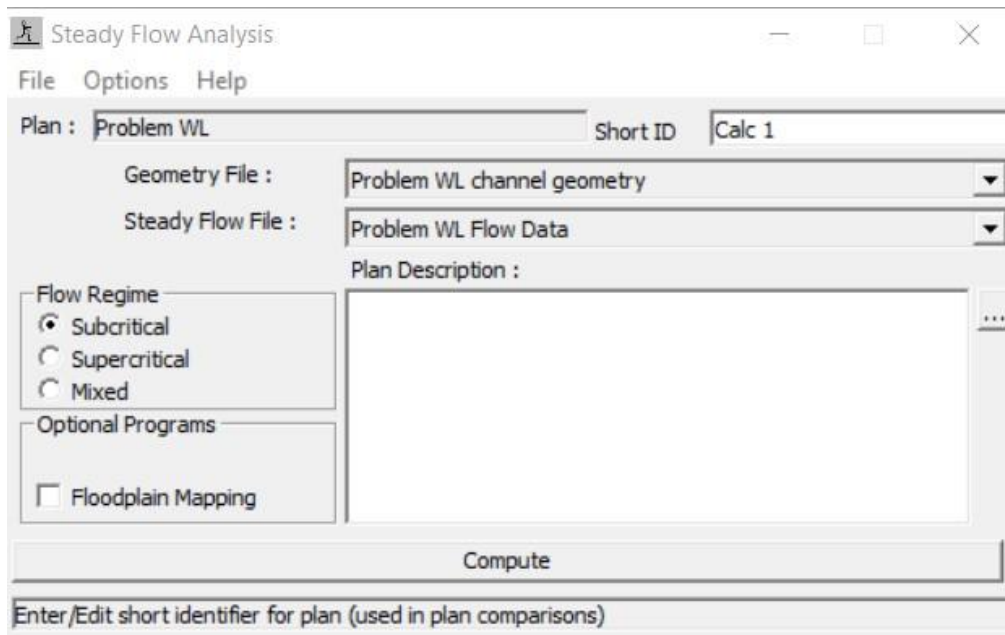
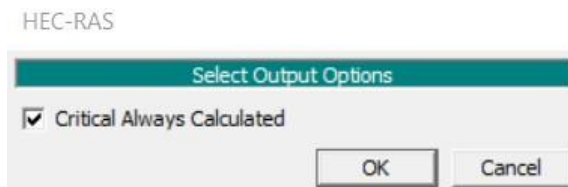


Figure 13: Steady Flow Analysis

Calculate critical depth for each cross section:

Activate **Options – Critical Depth Output Option**



Button **COMPUTE**

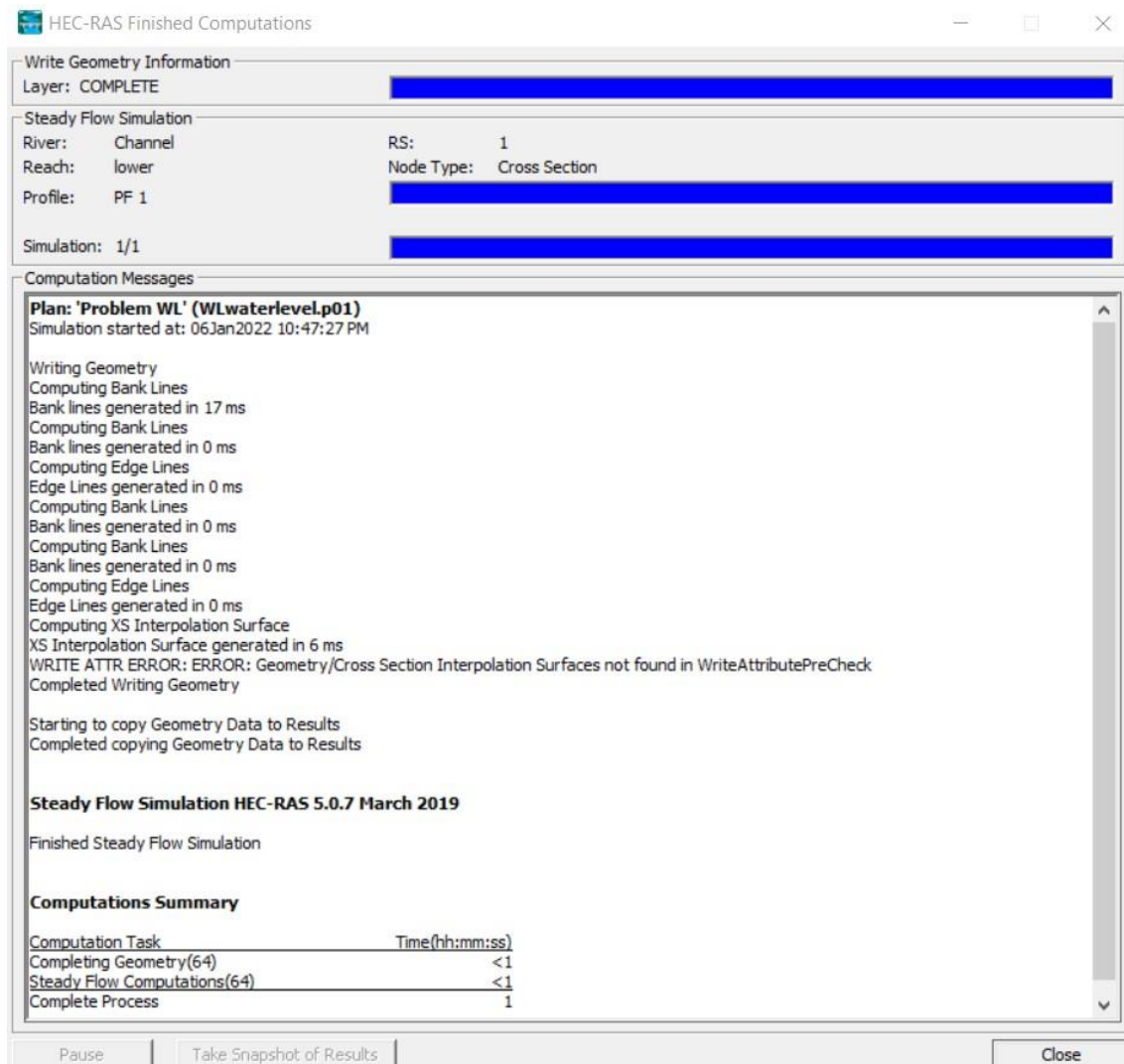


Figure 14: Display upon successful calculations

Button **Close**

View the error messages after calculations have been completed:

View – Summary, Err, Warn, Notes...

Warning: This may indicate the need for additional cross sections

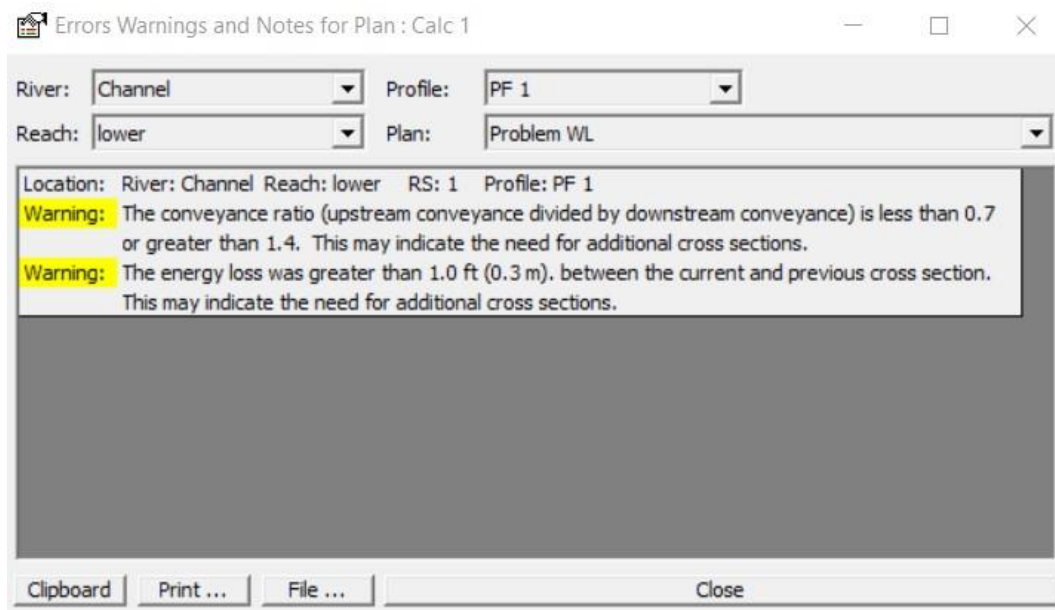


Figure 15: Display of errors, warnings and notes for the calculations

Button **Close**

Interpolate additional cross sections between the two existing cross sections

Interpolation of cross sections

Edit – Geometric Data

Tools – XS Interpolation – Within a Reach

Maximum Distance between XS's 1,225 m: adds an interpolated cross section every 1,225 m along the entire length (11,025 m) of the channel

XS Interpolation by Reach

River: Channel

Reach: lower

Upstream Riv Sta: (All RS)

Downstream Riv Sta: (All RS)

Maximum Distance between XS's: 1225

Cut Line GIS Coordinates

☒ Linearly interpolate cut lines from bounding XS's
(only available when bounding XS's are Georeferenced)

☐ Generate for display as perpendicular segments to reach invert
(will be repositioned as cross section data is changed)

Decimal places in interpolated Sta/Elev: 0.000

Delete Interpolated XS's Interpolate XS's

Close

Figure 16: Cross section interpolation between River Station 0 and 1

Button **Interpolate XS's**

Button **Close**

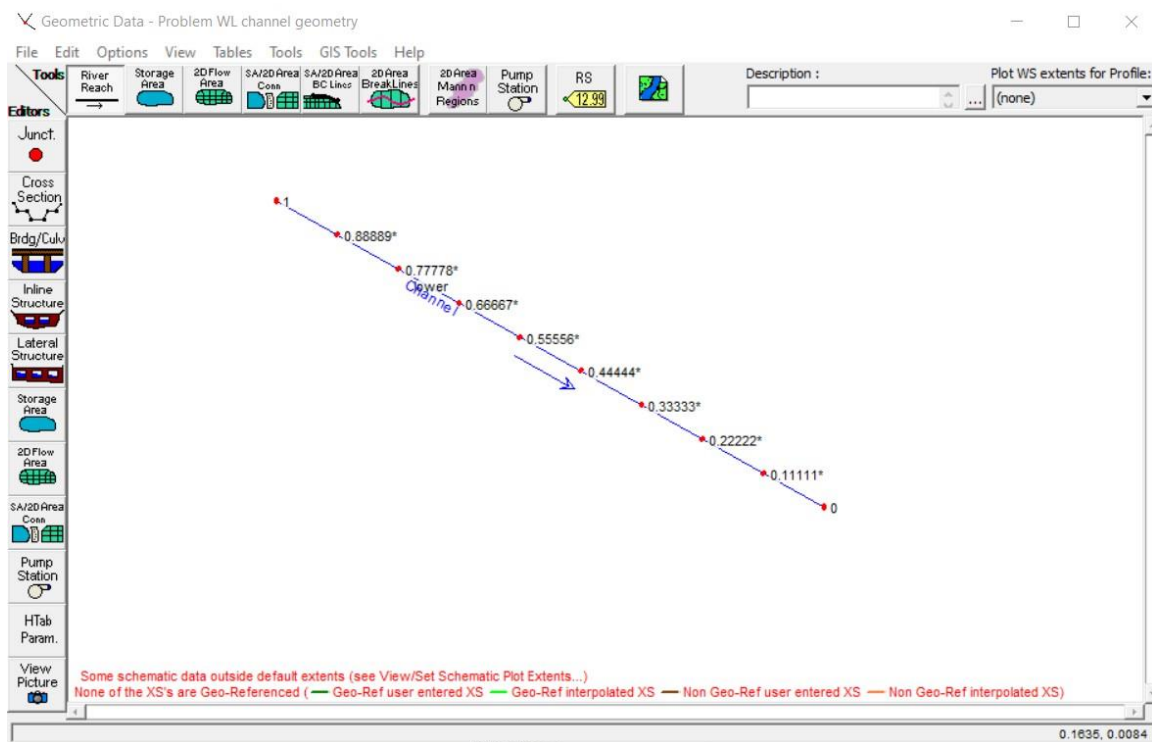
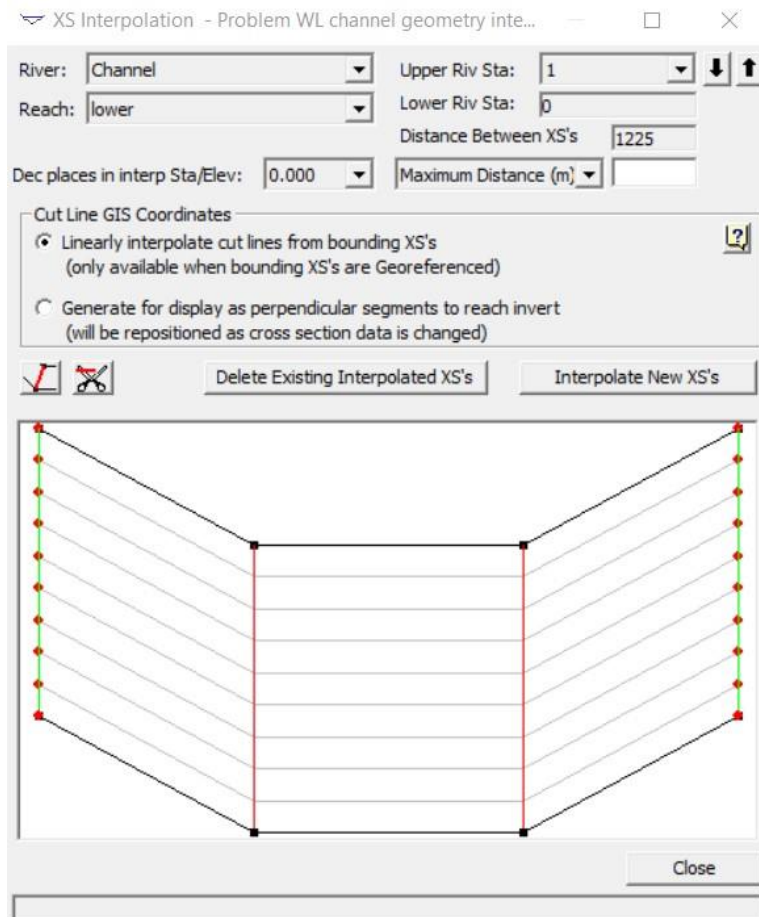


Figure 17: Channel with interpolated cross sections

Save new geometry: **File – Save Geometry Data As ...**
(**Title:** WL channel geometry interpolated)

View interpolation between cross sections:

Tools – XS Interpolation – Between 2 Xs's



New calculation

Run – Steady Flow Analysis

File - New Plan

(**Title:** WL interpolated, **Short Plan Identifier:** Calc2)

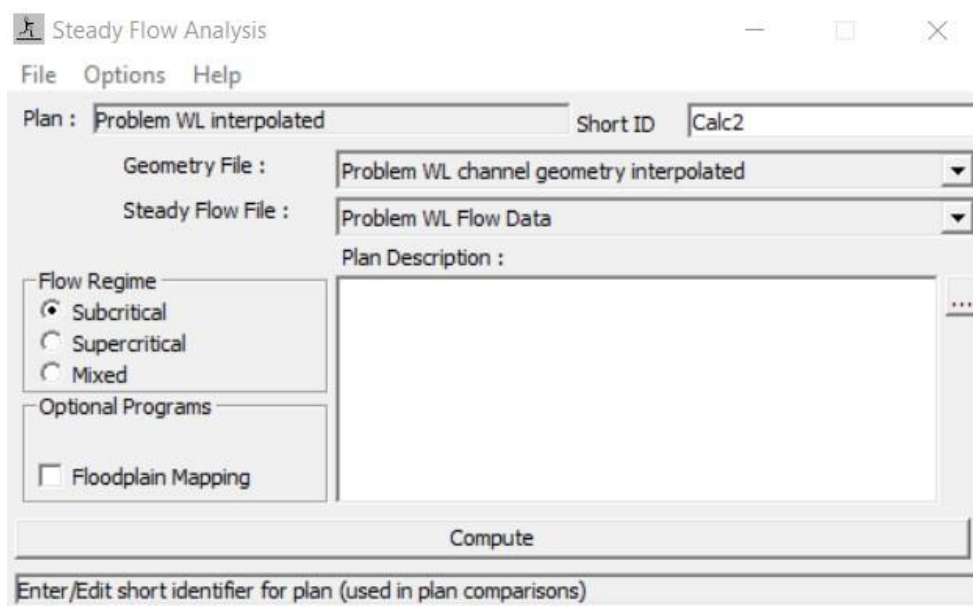
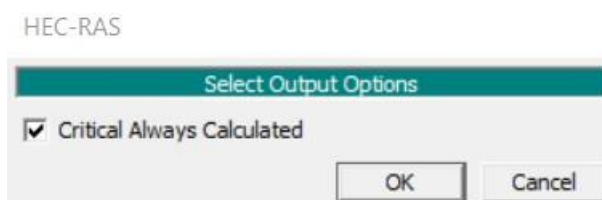


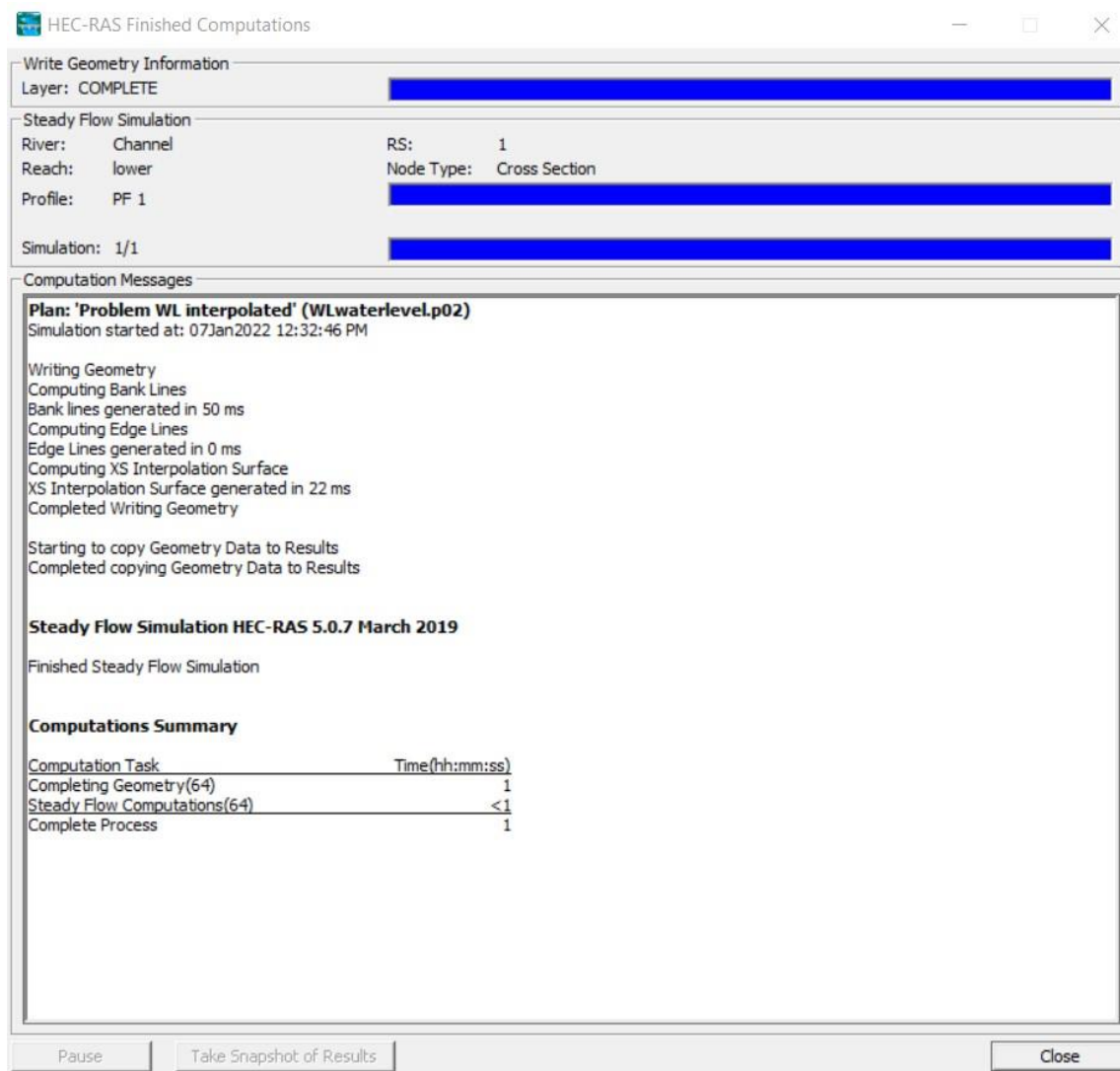
Figure 18: Steady Flow Analysis with new Geometry File

Calculate critical depth for each cross section:

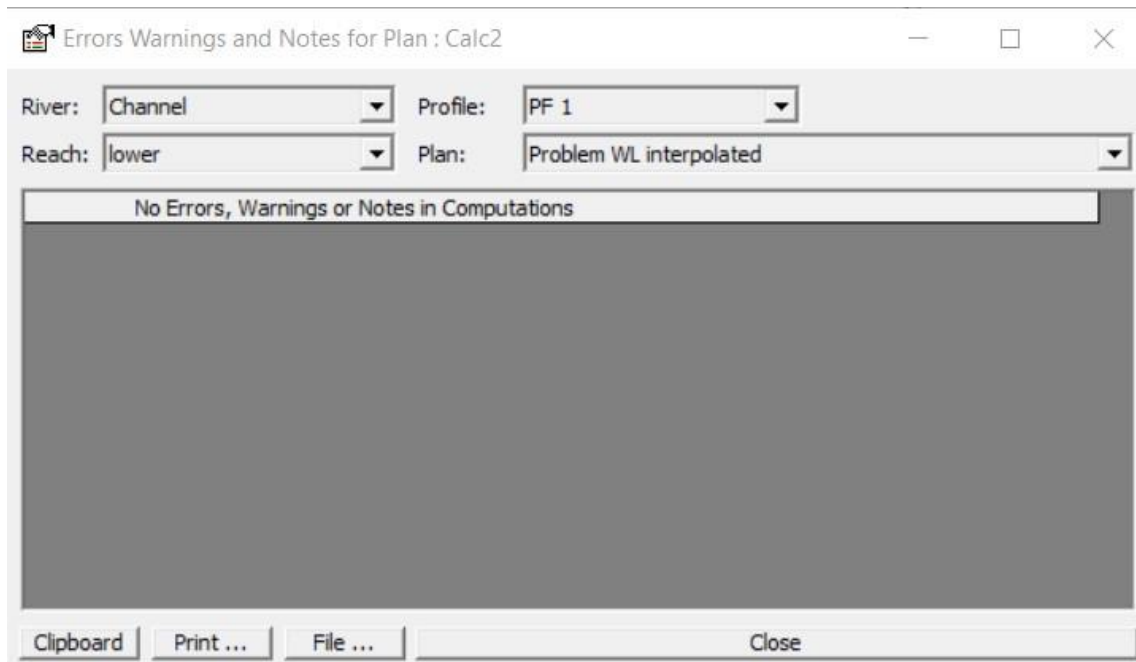
Activate **Options – Critical Depth Output Option**



Button **COMPUTE**



View – Summary, Err, Warn, Notes...



Display results

Graphic:

Longitudinal section view: **View – Water Surface Profiles**

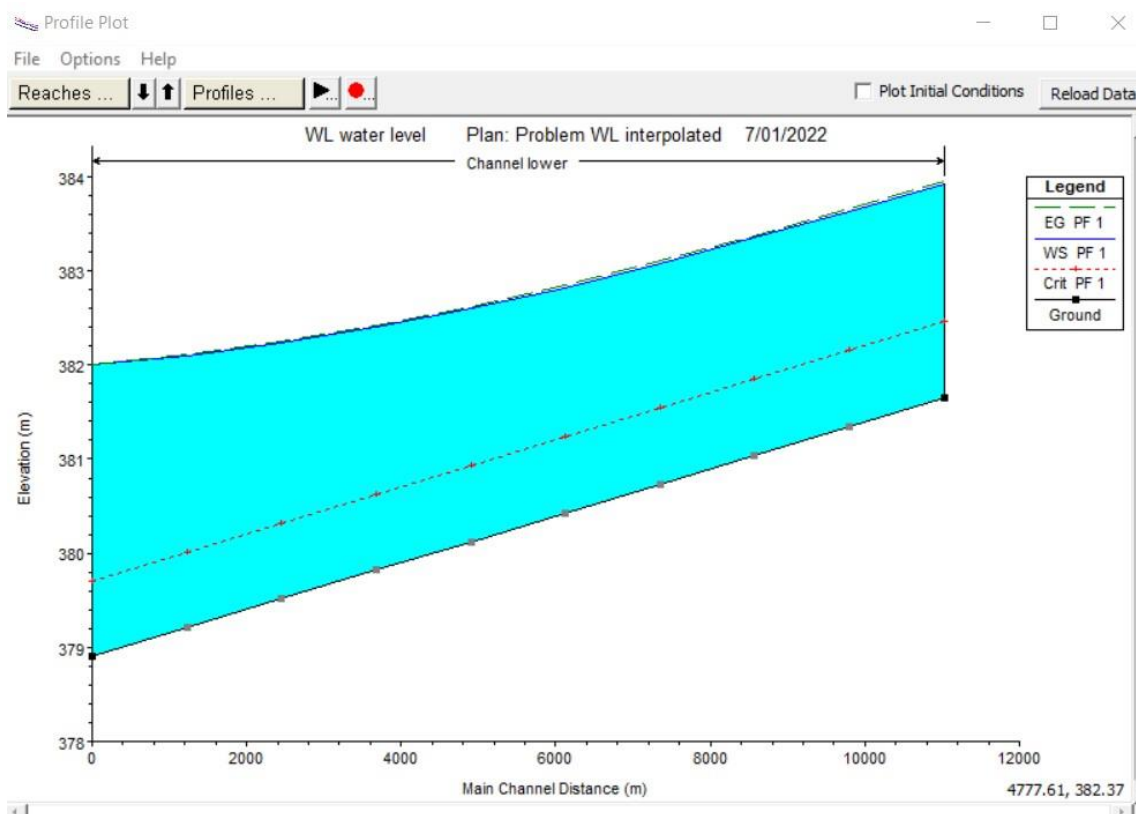


Figure 19: Water Surface Profiles (longitudinal profile)

Channel cross sections: **View – Cross Sections**

The buttons



can be used to view the different cross sections (also the interpolated ones)

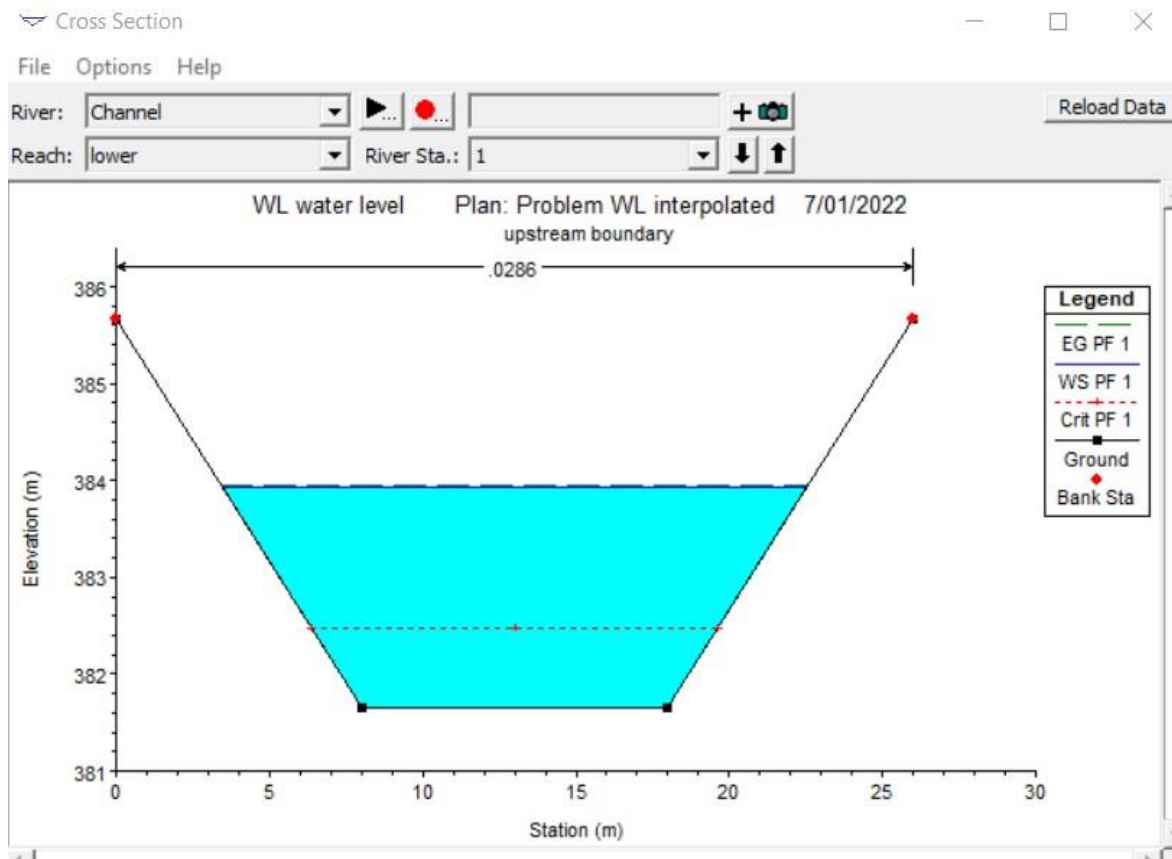


Figure 20: Cross Sections view (Cross sections of the channel)

Tables:

View – Detailed Output Tables (Type – activate Cross Section)

Cross Section Output

File Type Options Help

River: Channel Profile: PF 1

Reach: lower RS: 1 Plan: Calc2

Plan: Calc2 Channel lower RS: 1 Profile: PF 1

E.G. Elev (m)	383.96	Element	Left OB	Channel	Right OB
Vel Head (m)	0.03	Wt. n-Val.		0.029	
W.S. Elev (m)	383.93	Reach Len. (m)	0.00	1225.00	0.00
Crit W.S. (m)	382.47	Flow Area (m2)		33.03	
E.G. Slope (m/m)	0.000242	Area (m2)		33.03	
Q Total (m3/s)	25.00	Flow (m3/s)		25.00	
Top Width (m)	19.08	Top Width (m)		19.08	
Vel Total (m/s)	0.76	Avg. Vel. (m/s)		0.76	
Max Chl Dpth (m)	2.27	Hydr. Depth (m)		1.73	
Conv. Total (m3/s)	1606.6	Conv. (m3/s)		1606.6	
Length Wtd. (m)	1225.00	Wetted Per. (m)		20.16	
Min Ch El (m)	381.66	Shear (N/m2)		3.89	
Alpha	1.00	Stream Power (N/m s)		2.95	
Frctn Loss (m)	0.29	Cum Volume (1000 m3)		419.16	
C & E Loss (m)	0.00	Cum SA (1000 m2)		221.38	

Errors, Warnings and Notes

Select Profile

Figure 21: Detailed Output Table

View – Profile Summary Table

(Results shown in the Profile Summary Table: Discharge Q, minimum channel elevation, water level, critical water level, energy line, energy line slope, flow velocity, flow area, top width, Froude number)

Profile Output Table - Standard Table 1

File Options Std. Tables Locations Help

HEC-RAS Plan: Calc2 River: Channel Reach: lower Profile: PF 1

Reload Data

Reach	River Sta	Profile	Q Total (m3/s)	Min Ch El (m)	W.S. Elev (m)	Crit W.S. (m)	E.G. Elev (m)	E.G. Slope (m/m)	Vel Chnl (m/s)	Flow Area (m2)	Top Width (m)	Froude # Chl
lower	1	PF 1	25.00	381.66	383.93	382.47	383.96	0.000242	0.76	33.03	19.08	0.18
lower	0.88889*	PF 1	25.00	381.35	383.63	382.16	383.66	0.000237	0.75	33.28	19.14	0.18
lower	0.77778*	PF 1	25.00	381.04	383.35	381.85	383.38	0.000229	0.74	33.69	19.22	0.18
lower	0.66667*	PF 1	25.00	380.74	383.08	381.55	383.10	0.000216	0.73	34.37	19.36	0.17
lower	0.55556*	PF 1	25.00	380.43	382.82	381.24	382.85	0.000199	0.71	35.39	19.57	0.17
lower	0.44444*	PF 1	25.00	380.12	382.60	380.94	382.62	0.000177	0.68	36.93	19.88	0.16
lower	0.33333*	PF 1	25.00	379.82	382.40	380.63	382.42	0.000151	0.64	39.10	20.32	0.15
lower	0.22222*	PF 1	25.00	379.51	382.23	380.32	382.25	0.000123	0.59	42.04	20.89	0.13
lower	0.11111*	PF 1	25.00	379.21	382.10	380.02	382.12	0.000097	0.55	45.74	21.59	0.12
lower	0	PF 1	25.00	378.90	382.00	379.71	382.01	0.000075	0.50	50.22	22.40	0.11

Total flow in cross section.

Figure 22: Profile Summary Table

Save the project: **File – Save Project**